

IntelliPDF – AI Exam Learning App

Feature Ideas and Proposed Learning Workflow

1. Project Concept

IntelliPDF is an AI-powered personalized exam learning assistant that converts study materials into mark-wise preparation, practice, evaluation, and revision content.

2. Core Features

1. Mark-wise Answer Generator

Generate answers according to exam marks: 1M, 2M, 5M, 10M/15M/16M. Each answer can be structured according to the expected depth.

2. Important Topic Identification

Analyze the uploaded study material and identify topics that deserve focused study. The system should present these as important study areas rather than claiming to predict the actual question paper.

3. AI Study Plan

Create a daily study plan using the uploaded material, exam date, and the student's available study time.

4. Weak Topic Detection

Use quiz performance to identify topics where the student makes mistakes and needs more practice.

5. Personalized Revision

Reduce repetition for strong topics and provide additional questions and explanations for weak topics.

6. Answer Evaluation

Evaluate a student's written answer against the source material, identify missing points and conceptual issues, and suggest improvements with source references.

7. Exam Simulation Mode

Generate a timed mock test using selected mark categories such as 1M, 2M, 5M, and 16M, then provide score and topic-wise performance analysis.

8. Tamil + English Learning

Provide explanations in simple English and Tamil, with the possibility of a Tanglish-style explanation for easier understanding.

9. Source Verification

Show the page or section supporting an AI-generated answer so the student can verify the information in the original document.

10. Last-Minute Revision Mode

Create a quick revision package containing important definitions, key concepts, short notes, and important study questions for rapid preparation.

3. Suggested Overall Workflow

- 1 Upload PDF + syllabus
- 2 AI understands and processes the content
- 3 Identify important topics
- 4 Generate mark-wise questions and answers
- 5 Practice through quizzes and mock exams
- 6 Evaluate the student's answers
- 7 Detect weak areas
- 8 Create personalized revision
- 9 Show source/page references for verification

4. Project Identity

“An AI-powered personalized exam learning assistant that converts study materials into mark-wise preparation, practice, evaluation, and revision.”

5. Existing IntelliPDF Foundation

The current project report already includes PDF processing, NLP preprocessing, semantic search, RAG question answering, summarization, exam mode, multilingual support, and source/page references. The features above can be added around this existing foundation.